Mathematical Foundations of Cryptography (2 0 1– 3)

Course Description:

- The course gives students an overview of Cryptography and the mathematics behind Cryptography
- This course provides an insight into different cryptographic algorithms, its practical aspects, advantages and disadvantages.

Course Outcomes:

CO1	Understanding the mathematics behind cryptography
CO2	Learn Symmetric key cryptography and the advantages and disadvantages, how to build stream ciphers and detect the weaknesses and attacks
CO3	Implementation of DES, AES Algorithm and the corresponding attacks existing on them
CO4	Public key Cryptography advantages as well as various existing algorithms are explored, their proofs and how to currently attack them if implemented incorrectly.
CO5	Understand the difference between Digital Signatures and MACs, as well as the different algorithms existing plus their corresponding weaknesses
CO6	Learn about Hash functions properties, how to construct strong hash function and history of hash functions, as well as constructing SHA-1

CO-PO Affinity Map

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	1	-	-	-	3
CO2	3	2	1	-	-	-	3
CO3	3	2	1	-	-	-	2
CO4	3	2	1	-	-	-	2
CO5	3	2	1	-	-	-	3
CAM	3	2	1	-	-	-	3

Syllabus:

Unit-1 Concepts of Number Theory: Number Theory, GCD, Euclidean algorithm, Extended Euclidean algorithm, prime numbers, congruence, how to solve congruence equations, Chinese remainder theorem, residue classes and complete residue systems, Euler Fermat theorem, primitive roots.

Unit-2 Symmetric Key Cryptographic Systems : Caesar and affine ciphers, mono-alphabetic substitutions, transposition, homophonic, Vigenere and Beaufort ciphers, one-time pad, product/iterated/block ciphers, DES and AES. Heavy discussion is given to the security of these ciphers, not only are they studied in an algorithm sense but their attacks and defences are also discussed, modes of operation: CBC, ECB.

Unit-3 Cryptanalysis of symmetric keys : Attack Models, Linear, Differential and various others such as meet-in-the-middle attack .PKCS- Concepts of PKCS, Diffie Hellman key-exchange protocol, RSA, Rabin and EL Gamal cryptosystems.

Unit-4 Stream Ciphers and Digital Signatures: Synchronous, Self-synchronizing attack ciphers, linear feedback shift registers, Digital Signatures-RSA, multiple RSA and ElGamal signatures, digital signature standard.

Unit-5 Hash Functions and MACs: Hash functions, the Merkle-Damgard construction, Message Authentication Codes, security of Hash functions, security weakness of MD4, MD5, SHA1,SHA2, construction of SHA3.

Unit-6 Basic elliptic curve cryptography: definition, mathematical formulation of them, elliptic curve cryptography and pairings, elliptic digital signature algorithm, advantages, implementation of elliptic curve cryptography, point addition, point doubling, elliptic diffie hellman key exchange.

Unit 7: Key exchange protocols, Kerberos, Certificates, Man-in-the-middle-attack, Public key infrastructures

Textbooks/References:

1. William Stallings “Cryptography and Network Security” Fifth Edition, Prentice Hall, 2011
2. Alfred J Menzes, Paul C Van Oorshot and Scott A. Vanstone “Handbook of Applied Cryptography”, CBC Press, 1996
3. Stein William. “Elementary number theory. Primes, congruence’s, and secrets.” A computational approach
4. Neal Koblitz “A course in Number Theory and Cryptography” Springer-Verlag 1994
5. ChristofPaar, “Understanding Cryptography”, Springer-Verlag-2010

AMRITA VISHWA VIDYAPEETHAM Amrita Online - Course Plan					
Week	**Topic**	**Learning Objectives**	**eLearning Content**	**Video Lectures**	**Assessments**
1	**Introduction to Cryptography**	To get an Introduction to Cryptography	9 Videos 1 Quiz	Introduction to Cryptography	Quiz1
				Symmetric Cryptography	
				Simple Symmetric Encryption	
		To understand about Symmetric Cryptography		Letter frequency analysis	
				Cryptanalysis	
				How many key bits are enough?	
		To get an idea of basic mathematics used in Cryptography		Divisibility	
				Prime Numbers	
				Greatest common divisors	
2	**Fundamental Theorem of Arithmetic**	To understand the fundamental theorem of Arithmetic	8 Videos 1 Quiz	Fundamental Theorem of Arithmetic	Quiz 2
				Congruences	
		To get an idea of different types of ciphers		Practice problems 1	
				Practice problems 2	
				Practice problems 3	
		To understand how to solve practical problems		Integer Rings	
				Shift Cipher	
				Affine Cipher	
3	**Stream Cipher**	To understand stream ciphers in detailed	8 Videos 1 Quiz	Stream ciphers part 1	Quiz 3
		To understand different types of random number generators		Stream ciphers part 2	
		To get an idea of practical		Random Number Generators	

		stream ciphers			
		To understand how Linear Feedback Shift Registers (LFSR) works and how its behaviour leads to known plain text attack		One Time Pad	
				Towards Practical Stream Ciphers	
				Shift Based Stream Cipher	
				Examples of LFSR	
				Known Plaintext Attack	
4	**DES Algorithm**	To get an idea of another stream cipher, Trivium	9 Videos 1 Quiz	Trivium	Quiz 4
		To understand DES algorithm in detail		More about Stream Ciphers	
		To get an idea of security of DES algorithm		Practice problems 4	
				Introducti on to DES	
				Overview of DES Algorithm	
				Internal Structure of DES	
				Key Schedule	
				Decryption	
				Security of DES	
5	**AES Algorithm**	To understand AES algorithm in detail	7 Videos 1 Quiz	AES	Quiz 5
	Galois Field	To get an idea about Galois Fields		Galois Fields	
		To learn about software and hardware implementation of AES		Extension Fields part 1	
				Extension Fields Part 2	
				Internal Structure of AES part 1	
				Internal Structure of AES part 2	
				Hardware and Software Implementation	
	Block Ciphers	To learn more about block	9Videos 1 Quiz	More about	Quiz 6

6		ciphers		block ciphers	
		To understand how to improve the security of block ciphers		ECB Weakness	
		To know about Meet -in -the - Middle attack and Key whitening		CBC Mode	
				OFB and CFB Modes	
				CTR and GCM Mode	
				Exhaustive Key Search Revisited	
				Double Encryption	
				Triple Encryption	
				Key Whitening	
7	**Public Key Cryptography**	To understand Public key cryptography and its practical aspects	8 Videos 1 Quiz	Introduction to Public Key Cryptography	Quiz 7
		To learn about the mathematics needed for Public Key Cryptography		Practical Aspects of Public Key Cryptography	
				Mathematics Needed for Public Key Cryptography	
				Eulers phi function and more	
				Practice problems 5	
				Practice problems 6	
				RSA	
				Fast exponentiation	
8	**More About RSA**	To learn more about RSA algorithm	6 Videos 1 Quiz	Speed up techniques for RSA	Quiz 8
		To understand Discrete logarithm problems.		Practice problems 7	
				Finding large prime numbers	

				RSA Padding Final RSA Discrete Logarithm Problems	
9	**Discrete Logarithm Problem**	To understand the security aspects of Discrete Logaritm Problem To learn about Diffi Helman Key Exchange and its security	7 Videos 1 Quiz	Discrete Logarithm in Prime Fields Practice problems 8 Attacks against the DLP Security of DHKE ElGamal Computational Aspects Practice problems 9	Quiz 9
10	**Elliptic Curve Cryptography**	To understand Eliptic Curve Cryptography in detail To get an idea of the implementation of Eliptic Curves	7 Videos 1 Quiz	Introduction to Elliptic Curves Group Operations on Elliptic Curves Building a DLP on Elliptic Curves Elliptic Curve Diffie Hellman Key Exchange Practice problems 10 Practice problems 11 Implementation	Quiz 10
11	**Introduction to Diigital Signatures**	To understand Digital signaturesin detail	7 Videos 1 Quiz	Introduction to Digital Signatures RSA Digital Signature Scheme RSA Padding ElGamal Digital Signature Scheme Elgamal Digital Signature Security	Quiz 11

				DSA	
				ECDSA	
12	**Hashing**	Students will get an idea of Hashing.	9 Videos 1 Quiz	Practice problems 12	Quiz 12
		To understand the security requirements needed for hash function		Practice problems 13	
		To understand the implementation of hash functions		Hash functions	
				Security requirement of Hash functions part 1	
				Security Requirement for Hash Functions part 2	
				Overview of hash functions	
				Hash functions from block ciphers	
				SHA-1 Implementation	
				Implementation	
13	**Message Authentication Code**	To understand Message Authentication Code in detail	7 Videos 1 Quiz	Practice Problems 14	Quiz 13
				Introduction to MACs	
				MACs from hash function	
				HMACs	
				CBC MACs	
				Practice Problems 15	
				More about MACs	
14	**Key Distribution**	Understand key establishment and key distribution problem	9 Videos 1 Quiz	Key Establishment	Quiz 14
		To get an idea about Certificate Authority and		Key Distribution Problem	
				Key Distribution	

		Digital Certificates		Symmetric	
				Kerberos	
				Man In the Middle Attack	
				Certificates	
				PKIs	
				CAs	
				Practice problems 16	

Evaluation Policy

Internal	Marks calculated from the 14 quizzes published in LMS	30%	100 %
External	End semester Theory Exam – 70Marks	70%	

Faculty Information

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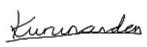

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